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Inventor(s)	ZHAO; Guanghui

SEATING ASSEMBLY, FOLDING MECHANISM, STORAGE DEVICE AND CHILD SAFETY SEAT

Abstract

The present disclosure provides a seating assembly for detachably connecting to a backrest of a child safety seat. The seating assembly includes a base; a seating body slidably connected to the base, the seating body being positioned at different positions with respect to the base; and a movable element located between the backrest and the seating body, wherein when the backrest is detached with respect to the seating body, the movable element is disengaged from the backrest and locked with the base to lock movement of the seating body with respect to the base. The present disclosure also provides a folding mechanism and a storage device. The present disclosure also a child safety seat including a seating assembly and/or a folding mechanism and/or a storage device.

Inventors:	**ZHAO; Guanghui (Dongguan, Guangdong, CN)**
Applicant:	**WONDERLAND SWITZERLAND AG** (Steinhausen, CH)
Family ID:	**1000008642423**
Assignee:	**WONDERLAND SWITZERLAND AG (Steinhausen, CH)**
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Background/Summary

[0001] This application is a National Stage application of PCT/EP2023/076271, entitled "SEATING ASSEMBLY, FOLDING MECHANISM, STORAGE DEVICE AND CHILD SAFETY SEAT" and filed on Sep. 22, 2023, which claims the benefit of Chinese Application No. 202211170356.0, filed on Sep. 23, 2022, Chinese Application No. 202211668176.5, filed on Dec. 23, 2022, and Chinese Application No. 202310008869.X, filed on Jan. 4, 2023, all of which are incorporated herein by reference in their entirety."

TECHNICAL FIELD

[0002] The present disclosure relates to a child safety seat, and specifically, relates to a seating assembly, a folding mechanism and a storage device of the child safety seat.

BACKGROUND

[0003] With increasing attention to the children's travel safety, a child safety seat has become an indispensable riding tool for a child.

SUMMARY

[0004] A first aspect of the present disclosure provides a child safety seat and a seating body thereof, so as to prevent uncomfortable seating feeling for children when the seat is in the usage mode without a backrest.

[0005] In one embodiment, the first aspect of the present disclosure provides a child safety seat. The child safety seat includes a seating assembly including a base and a seating body, the seating body being slidably connected to the base, the seating body being positioned at different positions with respect to the base; and a backrest detachably connecting to the seating body. The backrest includes a locking element, when the seating body does not slide to a preset position with respect to the base, the locking element is not capable of being unlocked, and when the seating body slides to the preset position with respect to the base, the locking element is capable of being unlocked.

[0006] In one embodiment, the seating assembly further includes a movable element located between the backrest and the seating body, wherein when the backrest is detached with respect to the seating body, the movable element is disengaged from the backrest and locked with the base to lock movement of the seating body with respect to the base.

[0007] In one embodiment, an engaging hole is formed in the base, and when the backrest is detached with respect to the seating body, the movable element is disengaged from the backrest and is engaged with the engaging hole.

[0008] In one embodiment, the backrest includes an abutting element, the movable element comprises a rotating rod rotatably connected to the seating body and an elastic element connected to the rotating rod, the rotating rod comprises a first end and a second end. When the backrest is not detached, the second end of the rotating rod is pressed by the backrest, so that the first end is opposite to the engaging hole and located outside the engaging hole, and the elastic element is configured to provide a force for moving the first end of the rotating rod toward the engaging hole.

[0009] In one embodiment, when the seating body slides to a preset position with respect to the base, the backrest is in a vertical state.

[0010] In one embodiment, a rear edge of the base is formed with an avoidance slot facing the backrest, and when the seating body slides to a preset position, the locking element is opposite to

the avoidance slot, so as to allow the backrest to be detached.
[0011] In one embodiment, a rear edge of the base is formed with an avoidance slot facing the backrest, when the seating body slides to a preset position, the locking element is unlocked through the avoidance slot, and when the seating body does not slide to the preset position, the locking element is not capable of being unlocked through the avoidance slot.
[0012] In one embodiment, the locking element comprises an engaging hook and an operating element rotatably provided on the backrest, the seating assembly further comprises an engaging rod fixed on the seating body, the engaging hook is fixedly connected to the operating element, the engaging hook has a hook part facing the engaging rod and hooking the engaging rod, the engaging hook slides with respect to the base along with the seating body, and the hook part is opposite to the avoidance slot when the seating body slides to the preset position.
[0013] In one embodiment, the seating body comprises a seat body, the seat body is formed with an accommodating groove for receiving the locking element, and the locking element is fixed to a bottom end of the backrest.
[0014] In one embodiment, the seating assembly comprises an adjustment assembly between the base and the seating body, the base comprises a positioning guide rail, and the positioning guide rail is provided with a plurality of positioning holes spaced apart from each other, and the seating body is slidably provided on the base and cooperates with respective one of the positioning holes through the adjustment assembly.
[0015] In one embodiment, the seating body is provided with a guide slot for accommodating the positioning guide rail, a length of the guide slot is greater than a length of the positioning guide rail, the seating body comprises a first abutting part, and the first abutting part is placed on a side of the guide slot away from the backrest to abut against the positioning guide rail when the seating body slides to a preset position.
[0016] In one embodiment, the seating body comprises a seat body, and the seat body comprises an upper shell and a lower shell connected to each other. The adjustment assembly comprises a positioning element connected between the upper shell and the lower shell, an elastic element connected to the positioning element, and an adjusting element connected to the positioning element. The positioning element is selectively placed in one of the positioning holes, and the adjusting element is movably connected to the lower shell.
[0017] In one embodiment, the positioning element comprises a positioning rod, a connecting element connected to the positioning rod, and a linking element connected between the adjusting element and the connecting element. The lower shell is formed with a first guide groove, an extending direction of the first guide groove is parallel to an axial direction of the positioning hole, the connecting element is movably placed in the first guide groove, the connecting element is formed with a receiving groove, and the elastic element is placed in the receiving groove and abuts against the connecting element.
[0018] In one embodiment, the seating body further comprises armrests on both sides of the seat body, and a connecting hole corresponding to the first guide groove is formed on an inner side of the armrest, and the positioning element passes through the positioning hole and is placed in the connecting hole.
[0019] In one embodiment, the seating body includes an armrest, a top portion of the armrest is formed with a first curved sliding slot and a second curved sliding slot communicated with each other, the backrest comprises a leaning part and side leaning parts on both sides of the leaning part, a sliding part protrudes from a bottom portion of each of the side leaning parts, the sliding part is slidably placed in the first curved sliding slot, a cross-section of each of the first curved sliding slot and the second curved sliding slot is matched with a cross-section of the sliding part.
[0020] The first aspect of the present disclosure also provides a seating assembly for detachably connecting to a backrest. The seating assembly includes a base; a seating body slidably connected to the base, the seating body being positioned at different positions with respect to the base; and a

movable element located between the backrest and the seating body, wherein when the backrest is detached with respect to the seating body, the movable element is disengaged from the backrest and locked with the base to lock movement of the seating body with respect to the base.

[0021] In one embodiment, an engaging hole is formed in the base, and when the backrest is detached with respect to the seating body, the movable element is disengaged from the backrest and is engaged with the engaging hole.

[0022] In one embodiment, an engaging hole is formed in the base, the movable element comprises a rotating rod rotatably connected to the seating body and an elastic element connected to the rotating rod, the rotating rod comprises a first end and a second end. When the backrest is not detached, the second end of the rotating rod is pressed by the backrest, so that the first end is opposite to the engaging hole and located outside the engaging hole, and the elastic element is configured to provide a force for moving the first end of the rotating rod toward the engaging hole.

[0023] In one embodiment, the seating assembly comprises an adjustment assembly between the base and the seating body, the base comprises a positioning guide rail, and the positioning guide rail is provided with a plurality of positioning holes spaced apart from each other, and the seating body is slidably provided on the base and cooperates with respective one of the positioning holes through the adjustment assembly.

[0024] In one embodiment, the backrest includes a locking element, the base is formed with an avoidance slot for cooperating with the locking element, the seating body comprises a seat body, the seat body is formed with an accommodating groove for receiving the locking element, and the avoidance slot at least partially overlaps with the accommodating groove.

[0025] In the child safety seat and the seating assembly, only when the seating body slides to the preset position, the backrest and the seating body may be unlocked therebetween, and the backrest may be detached from the seating body. In other words, the backrest may be detached from the seating body only when the seating body slides to the preset position with respect to the base. After the backrest is detached, the movable element is disconnected with the backrest and the base, the seating body will not be able to slide with respect to the base at this time. In this way, when the child safety seat is in the second mode with the backrest removed, the seating body will only be in the preset position, which can avoid misoperation by the user, thereby ensuring the comfort of the child in the second mode.

[0026] A second aspect of the present disclosure provides a child safety seat and a folding mechanism thereof. Specifically, the present disclosure provides a foldable child safety seat. More specifically, the present disclosure provides a folding mechanism of a child safety seat with simple structure, convenient operation and high safety level. Optionally, the present disclosure provides a child safety seat which may be folded through sliding connection between the seat part and the backrest.

[0027] In one embodiment of the second aspect provides a folding mechanism of a child safety seat for folding a backrest of the child safety seat with respect to the seat part. The folding mechanism includes a sliding groove disposed on one of the seat part and the backrest; a sliding rib disposed on the other of the seat part and the backrest; a locking structure operated to switch between a locking position and an unlocking position. The backrest is fixed with respect to the seat part and is in an unfolded state when the locking structure is in the locking position; and the backrest is foldable with respect to the seat part by means of sliding fit between the sliding groove and the sliding rib to be in a folded state when the locking structure is in the unlocking position.

[0028] In one embodiment, the sliding groove is disposed on each of both sides of the seat part, and the sliding rib is disposed on each of both sides of the backrest.

[0029] In an embodiment, the sliding groove is disposed on a top surface of an armrest at each of both sides of the seat part.

[0030] In one embodiment, the sliding rib is disposed on a bottom surface of each of two side wings of the backrest, and corresponds to the sliding groove.

[0031] In one embodiment, the armrest at each of both sides of the child safety seat part includes a front armrest and a rear armrest spaced apart from each other. In an unfolded state, the sliding rib is in a rear sliding groove of the rear armrest, and in a folded state, the sliding rib is in a front sliding groove of the front armrest.
[0032] In one embodiment, a virtual vertical plane is defined between the front armrest and the rear armrest. The virtual vertical plane is in a gap between the front armrest and the rear armrest in a front-rear direction and is perpendicular to a base. When the backrest of the child safety seat is locked with respect to the seat part, the backrest is located at a rear side of the virtual vertical plane; and when the backrest is unlocked with respect to the seat part, the backrest may move forward along the rear armrest and gradually pass through the virtual vertical plane, and then continue to move forward along the front armrest until the backrest is folded with respect to the seat. At this time, the entire backrest is located at a front side of the virtual vertical plane.
[0033] In one embodiment, the sliding rib and the sliding groove have complementary T-shaped structures, and a cross section of the sliding groove is in an inverted T shape.
[0034] In one embodiment, the locking structure is disposed on a lower side of the backrest. The folding mechanism further includes an engaging rod fixed to the child safety seat, and the locking structure is in the locking position by engagement with the engaging rod and is in the unlocking position by disengagement from the engaging rod.
[0035] In one embodiment, the locking structure includes an operating element at least partially exposed and disposed on the backrest; a locking hook assembly pivotally connected to a fixing rod of the backrest, and protrudes from a bottom side of the backrest, wherein the locking hook assembly is engaged with the engaging rod when the locking structure is in the locking position; and a connecting rod connected between the locking hook assembly and the operating element, wherein the connecting rod is fixedly connected with the locking hook assembly and movably connected with the operating element, wherein the locking hook assembly is capable of being disengaged from the engaging rod by operating the operating element.
[0036] In one embodiment, the locking hook assembly includes a first locking hook and a second locking hook. One end of each of the first locking hook and the second locking hook is connected with the operating element through a corresponding connecting rod of two connecting rods; the other ends of the first locking hook and the second locking hook are in hook shapes and opposite to each other; and middle parts of the first locking hook and the second locking hook are pivotally connected to the fixing rod of the backrest, respectively.
[0037] In one embodiment, one of the two connecting rods is located at an upper portion of the operating element, and the other of the two connecting rods is located at a lower portion of the operating element, the connecting rod at the lower portion is pivotally connected with the operating element, and the connecting rod at the upper portion is pivotably and slidably connected with the operating element.
[0038] In one embodiment, the locking structure includes two locking hook assemblies, and the two locking hook assemblies are disposed symmetrically with respect to the central axis of the backrest.
[0039] In an embodiment, the folding mechanism further comprises an elastic restoring element, one end of the elastic restoring element is fixed to a frame of the backrest, and the other end of the elastic restoring element is fixed to the locking structure, so as to apply a force to make the locking structure in the locking position.
[0040] In one embodiment, the other end of the elastic restoring element is fixed to the operating element, or the locking hook assembly, or the connecting rod.
[0041] In one embodiment, the backrest is removable from the seat part as being in a folded state.
[0042] The present disclosure also provides a child safety seat. The child safety seat includes a seat part; a backrest; and the folding mechanism according to any of the above embodiments, so that the backrest may be folded with respect to the seat part, and then fold the child safety seat.

[0043] In one embodiment, the child safety seat further includes a storage device detachably installed to the sliding groove of the seat part.
[0044] A third aspect of the present disclosure provides a storage device. The storage device is detachably installed on a seat part of a child safety seat. The storage device comprises: a connecting part slidably connected to the child safety seat in a front-rear direction; and an accommodating structure fixed on the connecting part for accommodating objects.
[0045] In one embodiment, the connecting part is slidable in the front-rear direction along a connecting structure of the child safety seat and is positioned at at least one position on the connecting structure.
[0046] In one embodiment, the connecting part comprises a first sliding part extending from one end of the connecting part and being on a lower side of the connecting part, to slidingly cooperate with a corresponding second sliding part provided on the connecting structure.
[0047] In one embodiment, the first sliding part is a sliding rib with an inverted T-shaped structure, and the second sliding part is a sliding groove that is complementary with the sliding rib in shape.
[0048] In one embodiment, the storage device further comprises a positioning element provided on the connecting part of the storage device. The storage device is positioned at different positions on the connection structure through the positioning element.
[0049] In one embodiment, the storage device further comprises a positioning element provided on the connecting part of the storage device. The positioning element is pivotably connected to the connecting part and has an engaged position in which the connecting structure is engaged and a disengaged position from which the connecting structure is disengaged.
[0050] In one embodiment, one end of the positioning element is pivotably connected to the connecting part, and the positioning element comprises a positioning protrusion on a lower side of the other end of the positioning element, for engaging with the child safety seat, so that the connecting part is positioned on the child safety seat.
[0051] In one embodiment, the positioning element further comprises an elastic arm having an arm-like structure extending from one end of the positioning element toward the other end of the positioning element, wherein a tail end of the elastic arm abuts against the connecting part to constantly provide a force to bias the positioning protrusion toward the child safety seat.
[0052] In one embodiment, the positioning element further comprises an operating part, and the operating part is pulled to overcome elastic rotation of the elastic arm, so that the positioning protrusion is disengaged with the child safety seat.
[0053] In one embodiment, the positioning element further comprises an operating part, and the operating part is operated to move the positioning element between the engaged position and the disengaged position.
[0054] The third aspect of the present disclosure provides a child safety seat. The child safety seat includes the storage device according to any one of the previous embodiments and a seat part. The storage device is placed on the seat part.
[0055] In one embodiment, a second sliding part of the child safety seat is provided on a top surface of an armrest on at least one side of the seat part.
[0056] In one embodiment, the second sliding part of the child safety seat is further provided with at least one positioning hole to engage with the positioning protrusion.
[0057] In one embodiment, the child safety seat further comprises a base, side covers and a bottom cover, wherein the side covers are fixedly connected to both sides of the child safety seat, and the bottom cover is fixedly connected between the base and the seat part.
[0058] In one embodiment, the child safety seat further comprises a backrest slidably connected to the seat part along the second sliding part of the seat to fold and unfold the backrest with respect to the seat part.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0059] The accompanying drawings are included to provide a further understanding of the present disclosure and are incorporated in and constitute a part of this specification, which illustrate embodiments of the present disclosure and together with the description below serve to explain the concept of the present disclosure.

[0060] FIG. **1** schematically shows a perspective view of a child safety seat according to an embodiment of a first aspect of the present disclosure.

[0061] FIG. **2** schematically shows an internal schematic view of a seating body in FIG. **1**.

[0062] FIG. **3** schematically shows a perspective view of a backrest in FIG. **1**.

[0063] FIG. **4** schematically shows an enlarged view of a part A in FIG. **2**.

[0064] FIG. **5** schematically shows a top view of the child safety seat in FIG. **1** in a preset position.

[0065] FIG. **6** schematically shows a cross-sectional view of the child safety seat taken along a line C-C in FIG. **5**.

[0066] FIG. **7** schematically shows a cross-sectional view of the child safety seat in FIG. **5** when it is in a non-preset position.

[0067] FIG. **8** schematically shows an enlarged view of a part B in FIG. **2**.

[0068] FIG. **9** schematically shows a cross-sectional view of a child safety seat according to another embodiment of the first aspect of the present disclosure.

[0069] FIG. **10** schematically shows a perspective view of a seating assembly in FIG. **1**.

[0070] FIG. **11** schematically shows a cross-sectional view of the child safety seat taken along a line D-D in FIG. **10**.

[0071] FIG. **12** schematically shows a perspective view of a child safety seat according to another embodiment of the first aspect of the present disclosure.

[0072] FIG. **13** schematically shows the child safety seat in FIG. **12** with the backrest removed.

[0073] FIG. **14** shows a perspective view of a child safety seat according to an embodiment of a second aspect of the present disclosure.

[0074] FIG. **15** shows a perspective view of a child safety seat in an unfolded state according to an embodiment of the second aspect of the present disclosure.

[0075] FIG. **16** shows a perspective view of the child safety seat of FIG. **15** when the locking structure is in an unlocking position and the backrest begins to be folded with respect to the seat part.

[0076] FIG. **17** shows a perspective view of the child safety seat of FIG. **16** that is further folded.

[0077] FIG. **18** shows a perspective view of the child safety seat of FIG. **16** in a fully unfolded state.

[0078] FIG. **19** shows a perspective view of the seat part of the child safety seat according to an embodiment of the second aspect of the present disclosure.

[0079] FIG. **20** shows a perspective view of the backrest of the child safety seat according to an embodiment of the second aspect of the present disclosure.

[0080] FIGS. **21***a* to **21***e* show views of the child safety seat when the locking structure is in a locking position according to an embodiment of the second aspect of the present disclosure, wherein FIG. **21***a* shows a front view of the child safety seat when the locking structure is in the locking position, FIG. **21***b* shows a sectional view taken along line E-E in FIG. **21***a*, FIG. **21***c* shows a partial enlarged view at a block F shown in FIG. **21***b*, FIG. **21***d* shows a perspective view of the child safety seat when the locking structure is in the locking position, and FIG. **21***e* shows a partial enlarged view of the locking structure shown in FIG. **21***d*.

[0081] FIGS. **22***a* to **22***e* show views of the child safety seat when the locking structure is in the unlocking position according to an embodiment of the second aspect of the present disclosure,

wherein FIG. **22***a* shows a front view of the child safety seat when the locking structure is in the unlocking position, FIG. **22***b* shows a sectional view taken along line G-G shown in FIG. **22***a*, FIG. **22***c* shows a partial enlarged view at a block H in FIG. **22***b*, FIG. **22***d* shows a perspective view of the child safety seat when the locking structure is in the unlocking position, and FIG. **22***e* shows the child safety seat at the locking structure in FIG. **22***d*.
[0082] FIG. **23** shows a perspective view of the child safety seat with a storage device installed on the seat part.
[0083] FIG. **24** schematically shows a perspective view of a seat according to an embodiment of a third aspect of the present disclosure.
[0084] FIG. **25** schematically shows a side view of a seat according to an embodiment of the third aspect of the present disclosure.
[0085] FIG. **26** schematically shows a perspective view of a storage device according to an embodiment of the third aspect of the present disclosure.
[0086] FIG. **27** schematically shows a perspective view of a seat part according to an embodiment of the third aspect of the present disclosure, in which a storage device is placed on a side of the seat part.
[0087] FIG. **28** schematically shows a perspective view of the seat part according to an embodiment of the third aspect of the present disclosure, in which the storage device on one side of the seat part is in an installed state, and the storage device on the other side of the seat part is being installed.
[0088] FIG. **29** schematically shows a perspective view of the seat part according to an embodiment of the third aspect of the present disclosure, in which the storage device is provided on each of both sides of the seat part.
[0089] FIG. **30** shows a top view of the seat part of FIG. **29**.
[0090] FIG. **31** shows a cross-sectional view of the seat part taken along a line J-J in
[0091] FIG. **30**.
[0092] FIG. **32** shows an enlarged view of a part K shown in FIG. **31**.
[0093] FIG. **33** shows a perspective view of the seat part of FIG. **29** from another angle.
[0094] FIG. **34** shows an enlarged view of a part Q shown in FIG. **33**.
TABLE-US-00001 List of reference numerals: seat 110 seating assembly 120 base 121 curved top surface 1211 positioning guide rail 1212 positioning hole 12121 engaging hole 12122 avoidance slot 1214 seating body 122 guide slot 1220 seat body 1221 upper shell 12211 lower shell 12212 installation port 1221h first abutting part 12213 first guide groove 12214 second guide groove 12215 accommodating groove 12216 armrest 1222 first curved sliding slot 12221 second curved sliding slot 12222 opening 12223 adjustment assembly 123 positioning element 1231 positioning rod 12311 connecting element 12312 linking element 12313 receiving groove 12314 first elastic element 1232 adjusting element 1233 movable element 124 rotating rod 1241 first end 12411 second end 12412 second elastic element 1242 rotating shaft 1243 object placement element 125 engaging rod 129 backrest 130 leaning part 131 side leaning part 132 sliding part 1321 abutting element 136 opening slot 1361 locking element 140, 140′ operating element 141 operating opening 1411 first fixing shaft 142 engaging hook 143 hook part 1431 second fixing shaft 145 elastic restoring element 147 child safety seat 21 seat part 210 base 2110 armrest 2120 front armrest 2120a rear armrest 2120b sliding groove 2130 front sliding groove 2130a rear sliding groove 2130b engaging rod 2140 locking hole 2150 storage device 2160 backrest 220 locking structure 2210 locking hook assembly 2211 first locking hook 2211a second locking hook 2211b connecting rod 2212 first connecting rod 2212a second connecting rod 2212b fixing rod 2213 elastic restoring element 2214 operating element 2215 connecting groove 2215a sliding rib 2230 cover plate 2240 headrest 230 virtual vertical plane P2 central axis M2 seat 31 backrest 310 seat part 320 housing 3210 side cover 3211 bottom cover 3212 armrest 3220 connecting structure 3221 second sliding part 32211 positioning hole 32212

abutting surface 32201 seat part body 3230 storage device 330 connecting part 3310 first sliding part 3311 positioning element 3320 first end 3320a second end 3320b elastic arm 3321 positioning protrusion 3322 operating part 3323 pin 3324 pivot point P3 accommodating structure 3330 base 340

DETAILED DESCRIPTION

[0095] The child safety seat configured to be installed in a vehicle can improve the safety of children while riding in the vehicle. To accommodate the children of different ages, the child safety seat typically offers two usage modes, that is, one usage mode with a backrest and the other usage mode without a backrest. In the usage mode with the backrest, a backrest angle may be adjusted to suit younger children. In the usage mode without the backrest, that is, after the backrest is removed from a seat part of the child safety seat, it may be used for older children. In the usage mode without the backrest, it is necessary to position the seat part at a specific position on a base. For example, a seating plane of the seat part is configured to be substantially parallel to a traveling direction of the vehicle, ensuring comfortable seating for the children. However, when the backrest of the existing child safety seat is removed, the seat part may be randomly adjusted with respect to the base, so that the seating plane may have different inclination angles. As such, it is difficult to ensure the comfortable seating for the children in the usage mode without the backrest.

[0096] Generally, the child safety seat is fixed on a back seat of the vehicle, occupying a large amount of space of the back seat of the vehicle. Most of the child safety seats currently on the market have a large size and are non-foldable. Placing the child safety seat in the back seat results in little left in the back seat of the vehicle or a limited space left for another passenger to sit or put some items thereon, to cause inconvenient for travelling. Moreover, when the child safety seat is not temporarily used, it is necessary to remove the entire child safety seat from the back seat of the vehicle to make room for passenger to sit or put items thereon; and when the child safety seat is used again, it is required to move the entire child safety seat back to the back seat of the vehicle and fix it thereon. It is not only cumbersome and inconvenient for operation, but also can increase transportation cost of the child safety seat. Furthermore, repeated loading and unloading can increase possibility of incorrect installation or unstable placement of the child safety seat, and a large space will be occupied to store the child safety seat that has been detached due to its large size, which can bring inconvenience to a user.

[0097] Furthermore, for sake of convenience, the children often place objects such as cups and toys on the child safety seat. The child safety seats are generally provided with storage cups that are usually detachable structures. However, traditional storage cups are usually detachably connected to the child safety seat in a vertical direction (i.e., an up-down direction). This means that if a child's foot accidentally kicks the storage cup from below, the storage cup may be skewed or accidentally detached, and potentially causing spills and safety hazards. In addition, the position of the storage cup in the existing child safety seat cannot be adjusted, which brings inconvenience to children of different sizes.

[0098] Hereinafter, exemplary embodiments of the present disclosure will be described in detail with reference to the accompanying drawings. While the present disclosure is susceptible to various modifications and alternative forms, specific embodiments thereof are shown by way of example in the accompanying drawings. However, the present disclosure should not be construed as limited to the embodiments set forth herein, but on the contrary, it will cover all modifications, equivalents and alternatives falling within the spirit and scope of the present disclosure.

[0099] The present disclosure will be described in detail with reference to the accompanying drawings.

[0100] Referring to FIGS. **1** and **2**, an embodiment of a first aspect of the present disclosure provides a child safety seat **110**. The child safety seat includes a seating assembly **120** and a backrest **130** detachably provided on the seating assembly **120**. The child safety seat **110** has a first mode in which the backrest **130** is provided and a second mode in which the backrest **130** is removed.

[0101] The seating assembly **120** includes a base **121**, a seating body **122**, and an adjustment assembly **123** between the base **121** and the seating body **122**. The seating body **122** is slidably connected to the base **121** and may move with respect to the base **121** so as to be positioned in different positions. Specifically, when a user operates the adjustment assembly **123**, a position of the seating body **122** with respect to the base **121** may be adjusted, thereby changing an inclination angle of a seating plane of the seating assembly **120** with respect to a horizontal reference plane.

[0102] Referring to FIGS. **1** and **2**, in one embodiment, the base **121** includes a curved top surface **1211** and two positioning guide rails **1212**. Each of the positioning guide rails **1212** is provided with a plurality of positioning holes **12121** spaced apart from each other. The seating body **122** is provided with two guide slots **1220** for accommodating the two positioning guide rails **1212**. A length of the guide slot **1220** is greater than a length of the positioning guide rail **1212**. The seating body **122** is slidably provided on the base **121** and may move forward and backward along the positioning guide rail **1212**.

[0103] As shown in FIG. **1**, the seating body **122** includes a seat body **1221** (which may be also referred to as “seat part”) and armrests **1222** on both sides of the seat body **1221**. The seat body **1221** includes an upper shell **12211** and a lower shell **12212** connected to each other. Referring to FIG. **2**, the guide slot **1220** is formed on the lower shell **12212**. The positioning guide rail **1212** of the base **121** is provided throughout the lower shell **12212** via the corresponding guide slot **1220**. In addition, the lower shell **12212** is also provided with a first abutting part **12213**. The first abutting part **12213** is provided at a side of the guide slot **1220** away from the backrest **130**. When the seating body **122** slides backward with respect to the base **121** to a preset position, the first abutting part **12213** is configured to abut against the positioning guide rail **1212**, thereby restricting further backward sliding of the seating body **122**.

[0104] That is, when the user cannot continue to slide the seating body **122** backward, the seating body **122** is in the preset position, that is, the backrest **130** is in a position where it may be detached from the seating body **122**. Similarly, the lower shell **12212** is also provided with a second abutting part (not shown). The second abutting part is provided at a side of the guide slot **1220** close to the backrest **130** and is configured to restrict further forward sliding of the seating body **122** when the seating body **122** slides forward to abut against the positioning guide rail **1212**. It can be understood that one or both of the first abutting part **12213** and the second abutting part may also be provided on the upper shell **12211**.

[0105] As shown in FIG. **2**, the adjustment assembly **123** is configured to be positioned with respect to different positioning holes **12121** to position the seating body **122** when the seating body **122** slides to different positions. In one embodiment, the adjustment assembly **123** includes a positioning element **1231** connected between the upper shell **12211** and the lower shell **12212**, a first elastic element **1232** connected to the positioning element **1231**, and an adjusting element **1233** connected to the positioning element **1231**. The positioning element **1231** is selectively placed in one of the positioning holes **12121**. The adjusting element **1233** is movably connected to the lower shell **12212** and may drive the positioning element **1231** to be disengaged from the positioning hole **12121** under the action of an external force. In one embodiment, a part of the adjusting element **1233** passes through a bottom portion of the lower shell **12212**, so that the user may operate the adjusting element **1233** to disengage the positioning element **1231** from the positioning hole **12121**. When the positioning element **1231** is disengaged from the positioning hole **12121**, the user may push the backrest **130**, so that the backrest **130** and the seating body **122** slide with respect to the base **121** to adjust the position of the seating body **122** on the base **121**. When the positioning rod **12311** moves to the other positioning hole **12121** and releases the external force exerted on the adjustment element **1233**, the positioning element **1231** may be positioned in the other positioning hole **12121** again through an elastic restoring force of the first elastic element **1232**. In this way, the seating body **122** and the backrest **130** are fixed to the base

121 again.

[0106] In one embodiment, the positioning element **1231** includes a positioning rod **12311**, a connecting element **12312** connected to the positioning rod **12311**, and a linking element **12313** connected between the adjusting element **1233** and the connecting element **12312**. In addition, two first guide grooves **12214** are provided on the lower shell **12212**, and an extending direction of each of the first guide grooves **12214** is parallel to an axial direction of the positioning hole **12121**. Each of the connecting elements **12312** is movably placed in the corresponding first guide groove **12214** to move toward or away from the positioning hole **12121** along the first guide groove **12214**, so that the positioning rod **12311** extends into or out of the positioning hole **12121**.

[0107] A connecting hole (not shown) corresponding to the first guide groove **12214** is formed on an inner side of the armrest **1222**. The positioning rod **12311** passes through the positioning hole **12121** and is placed in the connecting hole, so that the seating body **122** and the base **121** are more stably connected with each other. The connecting element **12312** is formed with a receiving groove **12314**, and the first elastic element **1232** is placed in the receiving groove **12314** and abuts against the connecting element **12312**, so that after the positioning rod **12311** is disengaged from the positioning hole **12121**, and in a process that the seating body **122** is sliding with respect to the base **121**, the positioning rod **12311** extends into the other positioning hole **12121** through an elastic force. Two sets of second guide grooves **12215** are also formed on the lower shell **12212**. The two linking elements **12313** of the two positioning elements **1231** are arranged to cross with each other, and each of the linking elements **2213** is movably placed in one set of second guide grooves **12215**. The two linking elements **12313** are connected to the adjusting element **1233**. The user can move the adjusting element **1233** away from the backrest **130** to drive the movement of the two linking elements **12313**, thereby driving the two positioning rods **12311** to be disengaged from their respective positioning holes **12121**. After the user releases the adjusting element **1233**, the backrest **130** and the seating body **122** slide under the pushing of the user until the positioning rod **12311** faces one of the positioning holes **12121**, and the positioning rod **12311** is inserted into the positioning hole **12121** under the action of the first elastic element **1232** to fix the seating body **122** to the base **121** again.

[0108] The backrest **130** is detachably connected to the seating body **122**. When the backrest **130** is connected to the seating body **122**, the backrest **130** may slide with respect to the base **121** along with the seating body **122**, and the child safety seat **110** may be used by younger children in the first mode in which the backrest **130** is provided.

[0109] Referring to FIG. **3**, the backrest **130** includes a locking element **140**. When the locking element **140** is unlocked, the backrest **130** may be detached from the seating body **122**. However, in this embodiment, the locking element **140** may be unlocked only when the seating body **122** is moved to the preset position with respect to the base **121**. That is, when the seating body **122** does not slide to the preset position with respect to the base **121**, the locking element **140** cannot be unlocked, and the backrest **130** cannot be detached from the seating body **122**. In this way, the backrest **130** can be prevented from being detached by the user when the seating body **122** is not adjusted to an appropriate position, thereby affecting the comfort of the children sitting thereon.

[0110] Referring to FIG. **3**, in one embodiment, the locking element **140** includes an engaging hook **143** and an operating element **141** rotatably provided on the backrest **130**. Specifically, referring to FIGS. **2** and **4**, the locking element **140** further includes a first fixing shaft **142** and a second fixing shaft **145** that are fixed on the backrest **130**. The operating element **141** and the engaging hook **143** are rotatably connected to the first fixing shaft **142**, and one end of the engaging hook **143** is connected to the operating element **141** through the second fixing shaft **145**. In this embodiment, an operating opening **1411** is provided at an upper end of the operating element **141** for the user to operate so as to rotate the operating element **141**.

[0111] When the seating body **122** does not slide to the preset position with respect to the base **121**, the locking element **140** locks the backrest **130** and the seating body **122** to prevent the backrest

130 from moving with respect to the seating body **122**. Referring to FIGS. **5** and **6**, when the seating body **122** slides with respect to the base **121** to the preset position shown in FIG. **6**, the backrest **130** and the seating body **122** may be unlocked by the locking element **140**, the backrest **130** may move with respect to the seating body **122**, so that the backrest **130** may be detached from the seating body **122**. At this time, the child safety seat **110** is in a second mode in which the backrest **130** is removed (see FIG. **10**), used for older children. As shown in FIG. **6**, when the seating body **122** slides to the preset position with respect to the base **121**, a center position of the seating body **122** is in a center position of the base **121**, and the backrest **130** is in a vertical state. With this arrangement, after the backrest **130** is removed, the child can sit comfortably in the seating body **122**.

[0112] Referring to FIG. **6**, the seating assembly **120** further includes an engaging rod **129** connected to the seating body **122**. Referring to FIGS. **4** and **6**, one end of the engaging hook **143** is linked with the operating element **141** through the second fixing shaft **145**, and the other end of the engaging hook **143** has a hook part **1431** facing the engaging rod **129** and hooking the engaging rod **129**.

[0113] In addition, in this embodiment, the base **121** is formed with an avoidance slot **1214** for matching with the locking element **140**. In detail, the avoidance slot **1214** is formed on a rear edge of the base **121**. When the seating body **122** slides to the preset position, the hook part **1431** of the engaging hook **143** corresponds to a position of the avoidance slot **1214**. In one embodiment, an orthographic projection of the hook part **1431** of the engaging hook **143** falls within the avoidance slot **1214**. When it is necessary to detach the backrest **130** from the seating body **122**, the operating element **141** may be rotated, and the hook part **1431** of the engaging hook **143** may be driven to rotate counterclockwise about the first fixing shaft **142**, so that the hook part **1431** is disengaged from the engaging rod **129**. By providing the avoidance slot **1214** on the base **121**, it can prevent the hook part **1431** from being interfered with and thus unable to be disengaged from the engaging rod **129** during rotation.

[0114] Referring to FIG. **4** again, in one embodiment, the locking element **140** further includes an elastic restoring element **147** connected to the second fixing shaft **145** with the operating element **141** for restoring the operating element **141** when the user releases the operating element **141**. Furthermore, one end of the elastic restoring element **147** is connected to the second fixing shaft **145**, and the other end of the elastic restoring element **147** is connected to the backrest **130**. When the external force applied to the operating element **141** is removed, an elastic force of the elastic restoring element **147** may restore the operating element **141** and drive the engaging hook **143** to be restored.

[0115] In another embodiment, the operating element **141** is connected to the backrest **130** in a transition fit. In this way, even if the user releases the operating element **141** after the hook part **1431** is disengaged from the engaging rod **129**, it will not automatically hook the engaging rod **129** again. In this way, when the hook part **1431** and the engaging rod **129** are just unlocked, and the backrest **130** and the seating body **122** are not separated, the backrest **130** and the seating body **122** may continue to maintain an unlocked state. On the other hand, when the backrest **130** is installed and it is necessary to lock the backrest **130** and the seating body **122**, the operating element **141** may be operated to rotate in an opposite direction, so that the hook part hooks the engaging rod **129**.

[0116] When the backrest **130** is not detached, the engaging hook **143** may slide with respect to the base **121** along with the seating body **122**. Therefore, the hook part **1431** may be movable along with the seat body **1221** with respect to the base **121**. Referring to FIG. **7**, when the seating body **122** is moved to the non-preset position with respect to the base **121**, the hook part **1431** of the engaging hook **143** will be misaligned with the avoidance slot **1214** and separated from each other. That is, an orthographic projection of the engaging hook **143** on the base **121** is located outside the avoidance groove **1214**.

[0117] In this case, the rotation of the engaging hook **143** may be interfered by the base **121**, and the user cannot rotate the operating element **141**, nor can the hook part **1431** be disengaged from the engaging rod **129**. Accordingly, when the seating body **122** is moved to the non-preset position with respect to the base **121**, the user cannot unlock the locking between the backrest **130** and the seating body **122** by rotating the operating element **141**, nor can the user detach the backrest **130** from the seating body **122**. In this way, it is possible to prevent the user from detaching the backrest **130** when the seating plane of the seating body **122** is still tilted with respect to the reference horizontal plane, thereby affecting the comfort of the occupant. In addition, in this embodiment, after the backrest **130** is detached from the seating body **122**, the movement of the seating body **122** will be locked, and the position of the seating body **122** can no longer be adjusted with respect to the base **121**.
[0118] Referring to FIGS. **2** and **8**, the seating assembly **120** further includes a movable element **124** connected between the backrest **130** and the seating body **122**. In one embodiment, engaging holes **12122** are formed on the positioning guide rail **1212** of the base **121**, and one end of the movable element **124** is disposed corresponding to the position of the engaging hole **12122**. In this embodiment, the engaging holes **12122** are opened facing the upper shell **12211**, but the present disclosure does not limit the position, number and direction of the engaging holes **12122**. In addition, the backrest **130** of this embodiment further includes an abutting element **136** located on the inner side, and an opening slot **1361** is formed at an end of the abutting element **136**.
[0119] In detail, the movable element **124** includes a rotating rod **1241** that is rotationally connected to the seating body **122** and a second elastic element **1242**. The rotating rod **1241** includes a first end **12411** and a second end **12412**. Referring to FIGS. **8** and **9**, the first end **12411** is opposite to the engaging hole **12122**, and the second end **12412** is placed in the opening slot **1361** and abuts against the abutting element **136**. Accordingly, when the backrest **130** is not detached, the second end **12412** of the rotating rod **1241** is pressed by the abutting element **136**, so that the first end **12411** is kept outside the engaging hole **12122**. It can be understood that the abutting element **136** may not be provided with the opening slot **1361**, and the second end **12412** directly abuts against a lower surface of the abutting element **136**. In this way, when the seating body **122** together with the backrest **130** moves with respect to the base **121**, the movable element **124** does not interfere with the movement of the seating body **122**.
[0120] Referring to FIG. **8**, the second elastic element **1242** is connected to the rotating rod **1241** and is configured to provide a force for moving the first end **12411** toward the engaging hole **12122** of the base **121**. In one embodiment, the upper shell **12211** is formed with a shaft hole (not shown), the rotating rod **1241** has a rotating shaft **1243** accommodated in the shaft hole and protruding from the rotating rod, and the second elastic element **1242** is a torsion spring sleeved on the rotating shaft **1243**. One end of the torsion spring abuts against the lower shell **12212**, and the other end of the torsion spring abuts against an upper surface of the rotating rod **1241**.
[0121] Referring to FIGS. **10** and **11**, when the backrest **130** and the seating body **122** move to a predetermined position, the backrest **130** may be detached. A top surface of the seat body **1221** is formed with an installation port **1221***h*, and the installation port **1221***h* is configured to install the abutting element **136** (in combination with FIG. **1**). After the backrest **130** is detached, it disengages from the abutment against the rotating rod **1241**, and a part of the rotating rod **1241** is exposed through the installation port **1221***h*. At this time, the second elastic element **1242** exerts a force on the first end **12411** of the rotating rod **1241**, so that the first end **12411** of the rotating rod **1241** is engaged with the engaging hole **12122**, and the seating body **122** is locked with the base **121**, preventing the seating body **122** from sliding with respect to the base **121**. In detail, after the backrest **130** is removed, the abutting element **136** is separated from the rotating rod **1241** and removes the force exerted on the second end **12412** of the rotating rod **1241**, and the second end **12412** of the rotating rod **1241** is exposed from the installation port **1221***h*. Through the elastic force of the second elastic element **1242**, the first end **12411** may be engaged in the engaging hole

12122, thereby locking the seating body **122** and the base **121**.

[0122] According to the child safety seat **110**, only when the seating body **122** slides to the preset position with respect to the base **121**, the backrest **130** and the seating body **122** may be unlocked therebetween, and the backrest **130** may be detached from the seating body **122**. In other words, the backrest **130** may be detached from the seating body **122** only when the seating body **122** slides to the preset position with respect to the base **121**. After the backrest **130** is detached, the backrest **130** is separated from the movable element **124** (i.e., does not abut against the movable element **124**) so that the movable element **124** is locked with the base **121**. At this time, the seating body **122** will not be able to slide with respect to the base **121**, and the user will no longer be able to adjust the position of the seating body **122** with respect to the base **121**. That is, with the mutual cooperation between the movable element **124** and the engaging hole **12122**, after the backrest **130** is detached from the seating body **122**, the movement of the seating body **122** is locked and cannot be moved anymore. In this way, when the child safety seat **110** is in the second mode with the backrest **130** removed, the seating body **122** will only be in the preset position, which can avoid misoperation by the user, thereby ensuring the comfort of the child in the second mode.

[0123] Referring to FIGS. **2** and **8** again, when the backrest **130** is re-installed on the seating body **122**, and in combination with FIG. **1**, the abutting element **136** of the backrest **130** extends into the installation port **1221***h* and presses against the movable element **124**, and the abutting element **136** presses down the second end **12412**. At the same time, the first end **12411** rotates away from the engaging hole **12122**. When the backrest **130** is completely installed on the seating body **122**, the first end **12411** is disengaged from the engaging hole **12122**. In this way, an angle between the seating body **122** and the backrest **130** may be adjusted with respect to the base **121**.

[0124] Referring to FIG. **10**, a first curved sliding slot **12221**, a second curved sliding slot **12222**, and an opening **12223** between the first curved sliding slot **12221** and the second curved sliding slot **12222** are formed on the top of the armrest **1222**. In addition, referring to FIG. **1**, the backrest **130** includes a leaning part **131** and side leaning parts **132** located on both sides of the leaning part **131**. A sliding part **1321** protrudes from the bottom of each of the side leaning parts **132**, and the sliding part **1321** is slidably placed in the first curved sliding slot **12221**. The seating assembly **120** further includes an object placement element **125**, which is detachably connected to the second curved sliding slot **12222** for placing objects. In one embodiment, a cross-section of each of the first curved sliding slot **12221** and the second curved sliding slot **12222** is matched with a cross-section of the sliding part **1321**. In detail, the cross-section of each of the first curved sliding slot **12221** and the second curved sliding slot **12222** is in an inverted T shape, and the cross-section of the sliding part **1321** is in an inverted T shape matching with that of the first curved sliding slot **12221**.

[0125] In this embodiment, when the object placement element **125** is detached from the second curved sliding slot **12222**, the sliding part **1321** sequentially slides from the first curved sliding slot **12221** to the opening **12223** and the second curved sliding slot **12222**, and extends out of the second curved sliding slot **12222**, thereby detaching the backrest **130** from the seating body **122**. In another embodiment, a cross-section of a part of the first curved sliding slot **12221** communicating with the opening **12223** is in an inverted T shape, and a cross-section of a part of the first curved sliding slot **12221** away from the opening **12223** is in a rectangle. A cross-section of the sliding part **1321** located at a front end of the bottom of the side leaning part **132** is in an inverted T shape, and a cross section of a part away from the front end of the bottom of the side leaning part **132** is in a rectangle. In this embodiment, when the sliding part **1321** slides along the first curved sliding slot **12221** until a part of the sliding part **1321** with an inverted T-shaped cross section is completely located in the opening **12223**, the user can lift the backrest **130** upward to disengage the sliding part **1321** from the first curved sliding slot **12221**, and detach the backrest **130** from the seating body **122**. In this embodiment, the user can detach the backrest **130** without detaching the object placement element **125**.

[0126] In the embodiment as mentioned above, when the seating body **122** slides to the preset position, the avoidance slot **1214** is located below the hook part **1431** to provide a reservation space for the engaging hook **143** to rotate. The user can rotate the operating element **141** to unlock the engaging hook **143** and the engaging rod **129**, and then push the backrest **130** forward along the first curved sliding slot **12221** to detach the backrest **130**. It can be understood that, during the detaching process, the user does not release the operating element **141** until the hook part **1431** cannot hook the engaging rod **129** when rotating in the opposite direction, so that the backrest **130** and the seating body **122** may also remain in the unlocked state after the operating element **141** is released. When the seating body **122** is in the non-preset position, the hook part **1431** is away from the avoidance slot **1214**, and the user cannot rotate the operating element **141** to unlock the hook part **1431** and the engaging rod **129**, and further cannot detach the backrest **130** from the seating body **122**.

[0127] As shown in FIG. **10**, in this embodiment, an accommodating groove **12216** is formed on a rear edge of the seat body **1221**, and the accommodating groove **12216** corresponds to the avoidance groove **1214** of the base **121**. When the seating body **122** slides to the preset position, the avoidance slot **1214** at least partially overlaps with the accommodating groove **12216**. Accordingly, after the backrest **130** is detached, the avoidance slot **1214** of the base **121** is exposed from the accommodating groove **12216**. When the backrest **130** is re-installed on the seating body **122**, the locking element **140** of the backrest **130** is placed in the accommodating groove **12216**, and the hook part **1431** of the engaging hook **143** engages with the engaging rod **129** of the seating body **122** (see FIG. **6**) to lock the backrest **130** and the seating body **122**.

[0128] Referring to FIGS. **12** and **13**, in another embodiment, a locking element **140′** is fixed on a bottom end of the backrest **130**, and the accommodating groove **12216** is formed on the rear edge of the seat body **1221** toward the base **121**. The locking element **140′** is placed in the accommodating groove **12216**, an end of the locking element **140′** slides along the curved top surface **1211** of the base **121** with the seat body **1221**. The accommodating groove **12216** may be a curved groove, and the locking element **140′** is attached between the curved groove and the curved top surface **1211**. The rear edge of the base **121** is formed with an avoidance slot (not shown) facing the backrest **130**, and the locking element **140′** is opposite to the avoidance slot when the seating body **122** is positioned at the preset position.

[0129] In the above embodiment, when the seating body **122** is positioned at the preset position and the user pushes the backrest **130** forward, the avoidance slot is located below the locking element **140′**, providing a reservation space for the locking element **140′** to rotate. Thus, when the backrest **130** slides along the first curved sliding slot **12221** to move the backrest **130** upward and rotate at the same time, the locking element **140′** may rotate away from the base **121** and disengage from the accommodating groove **12216**, thereby detaching the backrest **130** from the seating body **122**. When the seating body **122** is positioned at the non-preset position and the user pushes the backrest **130** forward, the locking element **140′** is located in the curved top surface **1211** and abuts against the curved top surface **1211**, thereby preventing the backrest **130** from sliding along the first curved sliding slot **12221**, and preventing the backrest **130** from being detached from the seating body **122**.

[0130] According to one embodiment, a seating assembly **120** as described above is provided for detachable connecting with the backrest **130**. The seating assembly **120** includes a base **121**, a seating body **122**, an adjustment assembly **123** connected between the base **121** and the seating body **122**, and a movable element **124** connected between the backrest **130** and the seating body **122**. The seating body **122** is slidably connected to the base **121** and may be positioned at different positions with respect to the base **121** by the user operating the adjustment assembly **123**. When the seating body **122** does not slide to the preset position with respect to the base **121**, the locking element **140**, **140′** is away from the corresponding avoidance slot. The rotation of the locking element **140**, **140′** will interfere with the base **121**, and the locking element **140**, **140′** will not be unlocked by the user, so that the backrest **130** cannot move with respect to the seating body **122**, and the backrest **130** cannot be detached from the seating body **122**. When the seating body **122** slides to the preset position with respect to the base **121**, the corresponding avoidance slot is located below the locking element **140**, **140**′, providing a reservation space for the locking element **140**, **140**′ to rotate, and the locking element **140**, **140**′ may be unlocked by the user, the backrest **130** may move with respect to the seating body **122**, and then the backrest **130** may be detached from the seating body **122**.

[0131] In addition, after the backrest **130** is detached, the movable element **124** is disengaged from the backrest **130** and locked with the base **121**, thereby locking the movement of the seating body **122**. The above-mentioned seating assembly **120** allows the backrest **130** to be detached only when the seating body **122** slides with respect to the base **121** to the preset position, and after the backrest **130** is detached, the seating body **122** and the base **121** are fixed, thereby maintaining the seating body **122** at the preset position to provide a comfortable riding experience for the children in the usage mode without the backrest provided.

[0132] Further, in order to solve the problems in the prior art, there is an urgent need for a foldable child safety seat with simple structure, convenient operation and high safety level, in particular to a folding mechanism of the child safety seat, so that the child safety seat can be folded when not in use, thereby reducing space occupancy rate and transportation cost.

[0133] An embodiment of a second aspect of the present disclosure provides a foldable child safety seat.

[0134] Referring to FIG. **14**, FIG. **14** shows a perspective view of a child safety seat according to an embodiment of the present disclosure. The child safety seat **1** includes a seat part **210** (similar to the "seat body **1221**" as described in the first aspect of the present disclosure), a backrest **220** and a folding mechanism for folding the child safety seat. Specifically, the folding mechanism may fold a backrest **220** of the child safety seat **21** with respect to the seat part **210**, so that the entire seat may be folded. In addition, the child safety seat **21** may further include a headrest **230**, and the headrest **230** is disposed on the backrest **220** and may be adjusted in a longitudinal direction.

[0135] As shown in FIG. **14**, the folding mechanism of the child safety seat **21** of the present disclosure includes a sliding groove **2130**, a sliding rib **2230**, and a locking structure (the locking structure in FIG. **14** is covered by a cover plate **2240**, and only an operating element **2215** thereof is exposed, for the specific structure of the locking structure, please see FIGS. **21***d* and **22***d* and it will be described later). The sliding groove **2130** is disposed on one of the seat part **210** and the backrest **220**, and the sliding rib **2230** is disposed on the other of the seat part **210** and the backrest **220**. For convenience of explanation, an example that the sliding groove **2130** is disposed on the seat part **210** and the sliding rib **2230** is disposed on the backrest **220** is taken, but the present disclosure is not limited thereto.

[0136] In one embodiment, the sliding groove **2130** is disposed on each of both sides of the seat part **210**, and the sliding rib **2230** is disposed on each of both sides of the backrest **220** (for example, on a side wing at each side of the backrest **220**). Specifically, the sliding groove **2130** is disposed on a top surface of an armrest **2120** at each of both sides of the seat part **210**, while the sliding rib **2230** is disposed on a bottom surface of each of two side wings of the backrest **220** and corresponds to the sliding groove **2130** on the seat part **210**, for example, a position of the sliding rib **2230** corresponds to a position of the sliding groove **2130** in a lateral direction, for cooperation and connection with each other. In addition, the sliding rib **2230** and the sliding groove **2130** have complementary structure, such as complementary T shapes, i.e., the sliding rib **2230** and the sliding groove **2130** have corresponding structures that are complementary to each other, wherein a cross section of the sliding groove **2130** has an inverted T shape. However, the present disclosure is not limited thereto, that is, the sliding rib **2230** and the sliding groove **2130** may also have other shapes and may cooperate with each other and slide with respect to each other.

[0137] FIGS. **15** to **18** respectively show different stages (processes) of the child safety seat **21** from an unfolded state to a fully folded state according to an embodiment of the present disclosure.

FIG. **15** shows the child safety seat **21** before being folded (that is, in the unfolded state), FIGS. **16** and **17** show the child safety seat **21** during being folded, and FIG. **18** shows the child safety seat **21** in the fully folded state.

[0138] As shown in FIG. **15**, when the backrest **220** is locked with respect to the seat part **210** and in the unfolded state, the child safety seat **21** is in a use state, an angle of the backrest **220** with respect to the seat part **210** is greater than or equal to 90 degrees, and the backrest **220** and the seat part **210** cannot move relatively, to ensure the safety of riding.

[0139] Referring to FIGS. **16** to **18** again, when the backrest **220** is unlocked from the seat part **210**, a lower end of the backrest **220** may be separated from the seat part **210**, and the backrest **220** may entirely slide with respect to the seat part **210** by means of the sliding fit between the sliding groove **2130** and the sliding rib **2230**, so as to be folded with respect to the seat part **210**. Specifically, the sliding ribs **2230** on both sides of the backrest **220** may slide forward along the corresponding sliding grooves **2130** on both sides of the seat part **210**, so that the backrest **220** and the seat part **210** gradually approach to each other (as shown in FIGS. **16** and **17**) until the child safety seat **21** is completely folded to the folded state (as shown in FIG. **18**).

[0140] Through the cooperation between the sliding ribs **2230** on both sides of the backrest **220** and the sliding grooves **2130** on both sides of the seat part **210**, the backrest **220** can be stably folded when the backrest **220** slides with respect to the seat part **10**, and a stable connection can be provided when the child safety seat is in the unfolded state, so as to avoid the backrest **220** from shaking laterally with respect to the seat part **210**.

[0141] FIG. **19** shows a perspective view of the seat part **210** of the child safety seat **1** according to an embodiment of the present disclosure. As shown in FIG. **19**, the armrest **2120** on each of both sides of the seat part **210** may include a front armrest **2120***a* and a rear armrest **2120***b* spaced apart from each other. In the case that a three-point safety belt of the vehicle is used for the child safety seat, the safety belt may pass through a gap between the front armrest **2120***a* and the rear armrest **2120***b*, to fix and protect the children sitting in the child safety seat. For a child safety seat in which a three-point seat belt is not used, the front armrest **2120***a* and the rear armrest **2120***b* may be integrated without the gap.

[0142] Referring to FIGS. **15** to **18**, a virtual vertical plane P**2** is defined between the front armrest **2120***a* and the rear armrest **2120***b*. The virtual vertical plane P**2** is located in the gap between the front armrest **2120***a* and the rear armrest **2120***b* in a front-rear direction and perpendicular to the base **2110**. As shown in FIG. **15**, when the child safety seat **21** is in the unfolded state, that is, when the backrest **220** is locked with respect to the seat part **210**, the sliding rib **2230** is located in the rear sliding groove **2130***b* of the rear armrest **2120***b*, that is, the backrest **220** is entirely located at a rear side of the virtual vertical plane P**2** (that is, one side of the rear armrest **2120***b*), and the angle of the backrest **220** with respect to the seat part **210** is greater than or equal to 90 degrees; while when the backrest **220** is unlocked with respect to the seat part **210** and the child safety seat **1** begins to be folded, as shown in FIGS. **16** and **17**, the backrest **220** begins to be folded (approach) with respect to the seat part **210**, and the sliding rib **2230** slides (which in turn drives the backrest **220** to move) forward along the rear sliding groove **2130***b* of the rear armrest **2120***b* and gradually passes through the virtual vertical plane P**2**, then crosses over the gap between the rear armrest **2120***b* and the front armrest **2120***a*, and slides into the front sliding groove **2130***a* of the front armrest **2120***a* (i.e., the angle of the backrest **220** with respect to the seat part **210** is less than 90 degrees) and continues to move forward along the front armrest **2120***a* until the sliding rib **2230** is completely located in the front sliding groove **2130***a* of the front armrest **2120***a*, and the backrest **220** is folded with respect to the seat part **210**, so that the child safety seat **21** is in the folded state (i.e., the fully folded state), at the same time, the entire backrest **220** is located at a front side of the virtual vertical plane P**2**.

[0143] In addition, a front end of the sliding groove **2130** on the armrest **2120** at each of both sides of the seat part **210** may be opened. In particular, a notch may be provided at an outer side on a front end of the front sliding groove **2130***a*. When the child safety seat **21** is completely folded (as shown in FIG. **18**), the sliding rib **2230** is located in the gap. At this time, the backrest **220** may be removed from the seat part **210** by pulling the side wings of the backrest **220** outward. After the backrest **220** is removed, the seat part **210** may be independently used as a “No-back Booster”. Furthermore, a rear side of the rear sliding groove **2130***b* is closed, to prevent the backrest **220** from sliding backward with respect to the seat part **210** when it reaches a position in which it is in the unfolded state, thereby further improving the connection reliability between the backrest **220** and the seat part **210** and the use safety of the child safety seat.

[0144] In one embodiment, as shown in FIG. **23**, the child safety seat **21** further includes a storage device **2160** detachably installed to the sliding groove **2130** of the seat part **210**. For example, a storage cup may be installed on the front side of the armrest **2120**, and the storage cup may be detachably inserted into the sliding groove **2130** to cover the sliding groove **2130**, so that the child safety seat **21** is more aesthetic, and foreign objects can be prevented from sliding into the sliding groove **2130** to block the sliding rib **2230** from sliding in the sliding groove **2130** or make the sliding uneven. When the seat needs to be folded, it is necessary to remove the storage device first.

[0145] Hereinafter, a locking structure **2210** for locking and unlocking between the backrest **220** and the seat part **210** will be described with reference to FIGS. **20** to **22***e* in combination with FIG. **19**. FIG. **20** shows a perspective view of the backrest **220** of the child safety seat **21** according to an embodiment of the present disclosure, FIGS. **21***a* to **21***d* respectively show views of the child safety seat **21** when the locking structure **2210** is in a locking position, and FIGS. **22***a* to **22***d* respectively show views of the child safety seat when the locking structure is in an unlocking position.

[0146] Referring to FIGS. **19**, **20**, **21***d* and **22***d*, the locking structure **2210** is disposed on a lower side of the backrest **220** and is operated to switch between the locking position and the unlocking position. The locking structure **2210** is in the locking position when it is engaged with an engaging rod **2140** fixed to the seat part **210**, so that the backrest **220** is locked with respect to the seat part **210**; while the locking structure **2210** is in the unlocking position when it is disengaged from the engaging rod **2140**, so that the backrest **220** can be moved and folded with respect to the seat part **210**.

[0147] Specifically, as shown in FIG. **21***d*, the locking structure **2210** includes a locking hook assembly **2211**, a connecting rod **2212** and an operating element **2215**. Referring to FIG. **14**, when the backrest **220** and the seat part **210** are assembled and locked together, the operating element **2215** is at least partially exposed to the backrest **220** for convenience of operation. The locking hook assembly **2211** and the connecting rod **2212** are located in the backrest **220** and covered by the cover plate **2240**, which can prevent dust or foreign objects from entering and prevent the users or children from accidentally putting their hands into the backrest and being injured. Meanwhile, when maintenance or service is needed, the cover plate **2240** may be opened to check whether the internal structure is abnormal, so that maintenance and operation costs can be saved.

[0148] As shown in FIGS. **20**, **21***d* and **22***d*, the locking hook assembly **2211** is pivotally connected to the fixing rod **2213** of the backrest **220**, and protrudes from a bottom side of the backrest **220** and into the seat part **210** through the locking hole **2150** on an upper surface of the seat part **210**, so that when the locking structure **2210** is in the locking position, the locking hook assembly **2211** is engaged with the locking rod **2140** fixed in the seat part **210**. The connecting rod **2212** is connected between the locking hook assembly **2211** and the operating element **2215**, wherein the connecting rod **2212** is fixedly connected with the locking hook assembly **2211** and movably connected with the operating element **2215**, so that the locking hook assembly **2211** may be disengaged from the engaging rod **2140** by operating the operating element **2215**.

[0149] In one embodiment, as shown in FIGS. **21***c* to **21***e*, the locking hook assembly **2211** includes a first locking hook **2211***a* and a second locking hook **2211***b*, wherein one end of each of the first locking hook **2211***a* and the second locking hook **2211***b* is connected with the operating element **2215** through a corresponding one of two connecting rods **2212** (i.e., a first connecting rod

2212*a* corresponding to the first locking hook **2211***a* and a first connecting rod **2212***b* corresponding to the second locking hook **2211***b*). The other end of each of the first locking hook **2211***a* and the second locking hook **2211***b* is in a hook shape, and the hook shapes of the other ends are opposite to each other. For example, the hook shape is formed as a pair of pliers as shown in FIG. **21***c*, and middle parts of the first locking hook **2211***a* and the second locking hook **2211***b* are pivotally connected to the fixing rod **2213**, respectively.

[0150] In addition, two connecting rods **2212** (i.e., a first connecting rod **2212***a* and a second connecting rod **2212***b*) are disposed up and down with respect to the operating element **2215**, that is, one of the two connecting rods **2212** is disposed at an upper portion of the operating element **2215**, and the other of the two connecting rods **2212** is disposed at a lower portion of the operating element **2215**. For example, as shown in FIGS. **21***e* and **22***e*, the connecting rod **2212***a* at the lower portion is pivotally connected with the operating element **2215**, while the connecting rod **2212***b* at the upper portion is pivotally and slidably connected with the operating element **2215**, so that the linkage relationship between the operating element **2215** and the locking hook assembly **2211** can be realized, and the locking hook assembly **2211** is smoothly driven to switch between the locking position and the unlocking position through the operating element **2215**.

[0151] In one embodiment, as shown in FIG. **21***d*, the locking structure **2210** includes two locking hook assemblies **2211** that are symmetrically disposed with respect to a central axis M**2** of the backrest **220**. The operating element **2215** may be disposed at the central axis of the backrest **220**, and the connecting rod **2212** penetrates into the operating element **2215** and has two ends respectively connected to two locking hook assemblies **2211** that are symmetrically disposed in a longitudinal direction, so that the two locking hook assemblies **2211** may move synchronously.

[0152] In addition, as shown in FIGS. **21***c* and **22***c*, the folding mechanism of the present disclosure further includes an elastic restoring element **2214**. One end of the elastic restoring element **2214** is fixed to a frame of the backrest **220**, and the other end of the elastic restoring element **2214** is fixed to the locking structure **2210** (for example, the operating element **2215**, the locking hook assembly **2211** or the connecting rod **2212**), to apply a force to make the locking structure **2210** in the locking position, so that the locking between the backrest **220** and the seat part **210** may be more stable and reliable. For example, the elastic restoring element **2214** may be a restoring tension spring, and one end of the restoring tension spring is disposed on one of the locking hooks or the connecting rods of the locking hook assembly to apply a force to pull the locking hook or the connecting rod to the locking position; or alternatively, the elastic restoring element **2214** may be a restoring compression spring, and one end of the restoring compression spring is provided on the operating element **2215** to apply a force to press the operating element back to its initial state (i.e., the locking structure is in the locking position). Hereinafter, an example will be described in which the elastic restoring element **2214** is disposed on the second locking hook **2211***b* or the second connecting rod **2212***b* connected with the second locking hook **2211***b*.

[0153] In order to facilitate the understanding of the working principle of the present disclosure, the switching process of the locking structure **2210** between the locking position and the unlocking position according to an embodiment of the present disclosure will be described in detail with reference to FIGS. **21***a* to **21***e* and **22***a* to **22***e*.

[0154] First, referring to FIGS. **21***a* to **21***e*, FIG. **21***a* shows a front view of the child safety seat **21** when the child safety seat **21** is in a use state and the locking structure **2210** is in the locking position, and FIG. **21***b* shows a sectional view taken along the line E-E in FIG. **21***a*, FIG. **21***c* shows a partial enlarged view at a block F shown in FIG. **21***b*, FIG. **21***d* shows a perspective view of the child safety seat when the locking structure **2210** is in the locking position, and FIG. **21***e* shows a partial enlarged view of the locking structure **2210** in FIG. **21***d*.

[0155] As shown in FIGS. **21***d* and **21***e*, the locking structure **2210** includes a locking hook assembly **2211**, a connecting rod **2212** and an operating element **2215**. The locking hook assembly **2211** includes a first locking hook **2211***a* and a second locking hook **2211***b*. The connecting rod

2212 includes a first connecting rod **2212***a* and a second connecting rod **2212***b*. The first connecting rod **2212***a* is connected with an upper end of the first locking hook **2211***a*. The second connecting rod **2212***b* is connected with an upper end of the second locking hook **2211***b*. A lower end of the first locking hook **2211***a* and a lower end of the second locking hook **2211***b* have hook structures opposite to each other. The first locking hook **2211***a* is located at a front side of the second locking hook **2211***b*, the first connecting rod **2212***a* is located at a lower side of the second connecting rod **2212***b*, and the first connecting rod **2212***a* is pivotally connected with the operating element **2215**. The second connecting rod **2212***b* is pivotally and slidably connected with the operating element **2215**. For example, as shown in FIG. **21***e*, the operating element **2215** has a connecting groove **2215***a* penetrating through the operating element **2215** along its width direction to be able to pivot and/or slide with respect to the operating element **2215**. The connecting groove **2215***a* has a length along a length direction of the operating element **2215**, and the second connecting rod **2212***b* may penetrate through the connecting groove **2215** and has a sliding stroke corresponding to the length of the connecting groove **2215***a*. Further, as shown in FIG. **21***c*, the locking structure **2210** further includes an elastic restoring element **2214**. One end of the elastic restoring element **2214** (for example, a restoring tension spring) is fixed to the frame of the backrest **220**, and the other end of the elastic restoring element **2214** is connected to the second locking hook **2211***b* or the second connecting rod **2212***b*. The elastic restoring element **2214** constantly applies a pulling force so that the first locking hook **2211***a* and the second locking hook **2211***b* are always engaged.

[0156] As shown in FIGS. **21***c*, **21***d* and **21***e*, when the locking structure **2210** is in the locking position, the first locking hook **2211***a* and the second locking hook **2211***b* are respectively engaged with the engaging rod **2140** fixed on the seat part **210** from two sides, and the second connecting rod **2212***b* is limited to one end of the sliding stroke of the connecting groove **2215***a*, for example, an end of the connecting groove **2215***a* close to the outside of the operating element **2215** as shown in FIG. **21***e*, to lock the backrest **220** to the seat part **210**. More specifically, as shown in FIG. **21***c*, a main part of the first locking hook **2211***a* and a main part of the second locking hook **2211***b* may be assembled into a shape similar to “X” in a mirror symmetrical manner. In the locking position, the first connecting rod **2212***a* connected to the upper end of the first locking hook **2211***a* is relatively away from the second connecting rod **2212***b* connected to the upper end of the second locking hook **2211***b* in the horizontal direction. However, the first locking hook **2211***a* and the second locking hook **2211***b* may have other shapes, and the present disclosure is not limited thereto.

[0157] Furthermore, referring to FIGS. **22***a* to **22***e*, FIG. **22***a* shows a front view when the locking structure **2210** is in the unlocking position and the child safety seat **21** has not been folded, FIG. **22***b* shows a sectional view taken along line G-G in FIG. **22***a*, FIG. **22***c* shows a partial enlarged view at a block H shown in FIG. **22***b*, FIG. **22***d* shows a perspective view of the child safety seat **1** when the locking structure **2210** is in the unlocking position, and FIG. **22***e* shows a partial enlarged view at the locking structure **2210** in FIG. **22***d*.

[0158] As shown in FIGS. **22***c*, **22***d* and **22***e*, when the backrest **220** needs to be unlocked and be folded with respect to the seat part **210**, the lower end of the operating element **2215** is pushed to pivot the lower end thereof inward and the upper end thereof outward, so that the second connecting rod **2212***b* slides in the connecting groove **2215***a* along the sliding stroke (as shown in FIG. **22***e*) and approaches the first connecting rod **2212***a* (as shown in FIG. **22***c*) to drive the first locking hook **2211***a* and the second locking hook **2211***b* (especially the lower end of the first locking hook **2211***a* and the lower end of the second locking hook **2211***b*) to pivot and move away from each other, so as to disengage from the engaging rod **2140**. At this time, the backrest **220** may be folded with respect to the seat part **210** through the cooperation of the sliding groove and the sliding rib. It can be understood that the locking structure **2210** may also be disposed in the seat part **210** and the engaging rod **2140** may be disposed on the backrest **220**.

[0159] When the operation (e.g., pushing) of the operating element **2215** is released, the elastic

restoring element **2214**, such as the restoring tension spring shown in FIG. **22***c*, pulls the second connecting rod **2212***b* (or the upper end of the second locking hook **2211***b*), in this way, the second connecting rod **2212***b* moves away from the first connecting rod **2212***a*, and the first locking hook **2211***a* and the second locking hook **2211***b* return to their initial positions, for example, their lower ends are close to each other. That is, when the backrest **220** are not movable with respect to the seat part **210**, the first locking hook **2211***a* and the second locking hook **2211***b* are re-engaged with the engaging rod **2140**, thereby returning to the locking position as shown in FIG. **21***c*.
[0160] Through the cooperation of the sliding groove and the sliding rib, and the locking and unlocking of the locking structure between the seat part and the backrest, the folding mechanism of the child safety seat can not only realize the folding and disassembling of the backrest with respect to the seat part, but also realize the stable connection in the lateral direction and the longitudinal direction, and improve the use safety and reliability. Moreover, the folding mechanism has simple structure and convenient operation, and the child safety seat can be folded when it is not in use, so that the space occupancy rate and the transportation cost can be reduced.
[0161] The child safety seat and the folding mechanism thereof have the following beneficial technical effects: [0162] The folding mechanism has a simple structure that is easy to operate and convenient for disassembly and assembly. The folding of the backrest with respect to the seat can be realized through the sliding fit between the seat part and the backrest, so that the volume of the folded child safety seat is small, and then the overall packaging can also be reduced, the transportation cost without occupying much space of the vehicle can be reduced, and it is not necessary for the users to disassemble and carry the entire child safety seat repeatedly to make room, and then the use convenience can be improved.
[0163] In addition, the connection between the backrest and the seat is more stable in the child safety seat of the present disclosure through the cooperation between the sliding ribs at both sides of the backrest (or the seat part) and the sliding grooves at both sides of the seat part (or the backrest), the lateral (i.e. in the left-right direction) shaking during traveling of the vehicle can be avoided; and the locking hook is engaged with the engaging rod fixed on the seat, so that the connection strength between the backrest and the seat part can be increased, the child safety seat has good strength when being used forward or backward, the stability of the child safety seat in the longitudinal direction can also be enhanced, and the use safety can be further ensured.
[0164] In addition, a third aspect of the present disclosure provides a seat.
[0165] Referring to FIGS. **24** and **25**, a perspective view and a side view of the seat **31** according to an embodiment of the present disclosure are schematically shown respectively.
[0166] Referring to FIG. **24**, the seat **31** of this embodiment includes a backrest **310**, a seat part **320**, and a storage device **330**. The backrest **310** is detachably installed on the seat part **320**. When the seat **31** is used, the backrest **310** may be provided substantially perpendicular to the seat part **320**, or an angle of the backrest **310** with respect to the seat part **320** may be adjusted within a certain range. When the seat **31** is not in use, the backrest **310** may be folded with respect to the seat part **320** (not shown) to reduce a volume of the entire seat and facilitate storage. The seat **31** may further include a base **340** disposed below the seat part **320** and configured to support and position the entire seat **31** on a vehicle seat. Optionally, the seat **31** of this embodiment is a child safety seat.
[0167] As shown in FIG. **24**, an armrest **3220** is provided on each of both sides of the seat part **320**. For example, the seat part **320** includes a seat part body **3230** and a housing **3210**. The housing **3210** includes two side covers **3211** and a bottom cover **3212**. Each of the two side covers **3211** is fixedly connected to an outer side of an upright portion on a corresponding one of left and right sides of the seat part body **3230**, and form an armrest **3220** with the corresponding upright portion. The bottom cover **3212** is fixedly connected between the base **340** and the seat part **320**. More specifically, as shown in FIG. **24**, the bottom cover **3212** is fixedly connected to the seat part body **3230** and the side covers **3211** on both sides respectively to form the appearance of the seat part

320.

[0168] In addition, a storage device **330** is provided on at least one armrest **3220** of the seat part **320**. The storage device **330** may be slidably connected to the seat **31**, such as its seat part **320**, in a front-rear direction (i.e., forward and rearward). For example, as shown in FIG. **24**, the storage device **330** may be installed on the armrest **3220** of the seat part **320** from the front of the seat in a direction toward the backrest **310**, or may be detached from the armrest **3220** in a direction away from the backrest **310**. More specifically, the storage device **330** may slide along an upper edge of the armrest **3220**. In one embodiment, a detachable storage device **330** is provided on each of the two armrests **3220**. For example, the storage device **330** may be a storage cup. The specific structure of the storage device **330** will be described below in combination with FIG. **26**.

[0169] Referring to FIG. **25**, when the storage device **330** is installed on the armrest **3220** of the seat part **320**, the storage device **330** may abut against an abutting surface **32201** on a front side of the armrest **3220** to prevent the storage device **330** from continuing to slide backward (i.e., the direction toward the backrest **310**).

[0170] Referring to FIGS. **26** and **27**, FIG. **26** schematically shows a perspective view of a storage device **330** according to an embodiment of the present disclosure; and FIG. **27** schematically shows a perspective view of the seat part **320** according to an embodiment of the present disclosure, wherein the armrest **3220** on one side of the seat part **320** is provided with the storage device **330**, while the armrest **3220** the other side of the seat part **320** is not provided with the storage device **330** to illustrate an internal structure at a connection between the seat part **320** and the storage device **330**. It should be noted that an embodiment in which the storage device **330** is disposed on the seat part **320** is shown, but the present disclosure is not limited thereto, that is, the storage device **330** may also be disposed at other positions of the seat.

[0171] As shown in FIGS. **26** and **27**, the storage device **330** includes a connecting part **3310** and an accommodating structure **3330**. The storage device **330** may be slidably connected to the seat **31** through the connecting part **3310**. For example, the storage device **330** may be slidably connected to the seat part **320** and slide with respect to the armrest **3220** of the seat part **320**, so that the storage device **330** can be easily installed and detached. The accommodating structure **3330** is fixed on the connecting part **3310** for accommodating objects. In one embodiment, the accommodating structure **3330** may be disposed at one end of the connecting part **3310** in an offset manner. For example, referring to FIG. **27**, when the storage device **330** is installed on the armrest **3220** of the seat part **320**, the accommodating structure **3330** may be located outside a front end of the connecting part **3310** (outside with respect to the seat part **320**).

[0172] Further, as shown in FIG. **26**, the connecting part **3310** includes a first sliding part **3311**. Correspondingly, as shown in FIG. **27**, the seat **31** (only the seat part **320** of the seat **31** is shown, for other parts of the seat **31**, please referring to FIG. **24**) is also provided with a connecting structure **3221**. In one embodiment, the first sliding part **3311** extends from one end of the connecting part **3310** and is located on a lower side of the connecting part **3310** for slidingly cooperating with the connecting structure **3221**. For example, the connecting structure **3221** may be provided on the seat part **320** (such as its armrest **3220**), or may be provided on any side surface of the backrest (not shown). The connecting part **3310** of the storage device **330** may slide forward and backward along the connecting structure **3221** of the seat **31** and be positioned at least one position on the connecting structure **3221**.

[0173] More specifically, the connecting structure **3221** includes a second sliding part **32211** and a positioning hole **32212** (the positioning hole **32212** will be further described below in combination with FIGS. **31** and **32**). The first sliding part **3311** is slidably cooperated with the corresponding second sliding part **32211** provided on the connecting structure **3221**. In one embodiment, the second sliding part **32211** is provided on a top surface of the armrest **3220** on at least one side of the seat part **320**. More specifically, as shown in FIG. **27**, the second sliding part **32211** may be formed between the upright portion on any side of the seat part body **3230** and the corresponding side cover **3211**. In one embodiment, the first sliding part **3311** may be a sliding rib with an inverted T-shaped structure (see FIG. **26**), and the second sliding part **32211** may be a sliding groove complementary with a shape of the sliding rib (see FIG. **27**), but the present disclosure is not limited thereto, and the first sliding part and the second sliding part may also have other shapes that slide and fit with each other.

[0174] In addition, in one embodiment, a structure that slidingly cooperated with the second sliding part **32211** may also be provided below the backrest of the seat (see FIG. **24**), so as to be slidable along the second sliding part **32211**, thereby realizing angle adjustment and folding of the backrest with respect to the seat part.

[0175] Referring to FIGS. **28** and **29**, a process of installing the storage device **330** to the seat part **320** is shown. Specifically, FIG. **28** schematically shows a perspective view of the seat part **320** according to an embodiment of the present disclosure, in which the storage device provided on one side of the seat part **320** is in an installed state, and the storage device provided on the other side of the seat part **320** is being installed; and FIG. **29** schematically shows a perspective view of the seat part **320** according to an embodiment of the present disclosure, in which the storage devices **330** on both sides of the seat part **320** are in the installed state.

[0176] As shown in FIGS. **28** and **29**, the connecting part **3310** of the storage device **330** may be installed to either side of the seat part **320** from the front end of the seat part **320** along the second sliding part **32211** provided on the armrest **3220**.

[0177] Through the cooperation of the first sliding part **3311** of the connecting part **3310** and the second sliding part **32211** on the armrest **3220**, the storage device **330** may be installed on the seat part **320** from the front of the seat part **320**, thereby simultaneously restricting the movement of the storage device **330** in a left-right direction and in an up-down direction of the seat part **320**, thereby ensuring the stability of the storage device **330** and the objects placed therein to avoid possible tilting or overturning of the storage device **330** when a child moves to the seat part **320** from one side of the seat and accidentally hits the storage device **330** sideways, or when a child sitting on the seat part **320** accidentally kicks the storage device **330** from below or sideways, thereby preventing the objects in the storage device **330** from spilling.

[0178] In addition, FIG. **30** shows a top view of the seat part **320** of FIG. **29** and the storage device **330** provided thereon. As shown in FIG. **30**, when the storage device **330** is installed on the armrest **3220** of the seat part **320**, the accommodating structure **3330** of the storage device **330** may be offset outward with respect to the seat part **320** at the front end of the entire seat part **320**, further reducing the possibility that legs or feet of the child sitting on the seat part **320** accidentally hits the accommodating structure **3330**.

[0179] Referring to FIGS. **31** and **32**, FIG. **31** shows a cross-sectional view of the seat part **320** taken along a line J-J of FIG. **30**, and FIG. **32** shows an enlarged view of a part Q shown in FIG. **31**.

[0180] As shown in FIGS. **31** and **32**, the storage device **330** further includes a positioning element **3320**, and the positioning element **3320** is provided on the connecting part **3310**. The storage device **330** may be positioned at different positions on the connecting structure (only the positioning holes **32212** of the connecting structure are shown in FIG. **32**, and the connecting structure will be described in detail later) through the positioning element **3320**. In one embodiment, the positioning element **3320** is pivotably connected to the connecting part **3310** of the storage device **330**, and has an engaged position (as shown in FIG. **31**) in which it is engaged with the connecting structure **3221** and a disengaged position (not shown) in which it is disengaged from the connecting structure **3221**. Specifically, when the storage device **330** is installed on the seat part **320**, the positioning element **3320** may be disposed between the connecting part **3310** of the storage device **330** and the seat part **320** to prevent sliding between the storage device **330** and the seat part **320** and position the connecting part **3310** on the seat portion **320**. For example, one end of the positioning element **3320** may be pivotably connected to the connecting part **3310** through a pin **2324**, for example, its connecting position is at a pivot point P**3** as shown in FIG. **32**. In combination with FIGS. **31** and **32**, it can be seen that one end of the positioning element **3320** that is pivotally connected to the connecting part **3310** (i.e., an end close to the pivot point P**3**) is further away from the accommodating structure **3330** of the storage device **330** than the other end of the positioning element **3320**.

[0181] In one embodiment, the positioning element **3320** further includes an elastic arm **3321**. The elastic arm **3321** has an arm-like structure extending from one end of the positioning element **3320** toward the other end of the positioning element **3320**. Specifically, as shown in FIGS. **31** and **32**, the arm-like structure of the elastic arm **3321** extends from a first end **3320***a* of the positioning element **3320** (that is, an end close to the pivot point P**3**) toward a second end **3320***b* of the positioning element **3320** (that is, an end away from the pivot point P**3**), and the arm-like structure is spaced apart from the second end **3320***b* with a certain distance. In addition, an end **2321***a* of the elastic arm **3321** abuts against an inner surface of the connecting part **3310** to bias a main part of the positioning element **3320** toward the seat part **320** therebelow, and then position the connecting part **3310** when the connecting part **3310** slides to an appropriate position of the seat (for example, the seat part **320** as shown in FIG. **31**).

[0182] In one embodiment, the positioning element **3320** further includes a positioning protrusion **3322**. The positioning protrusion **3322** is located on a lower side of the other end of the positioning element **3320** for engaging with the seat, so that the connecting part **3310** is positioned on the seat. More specifically, as shown in FIGS. **31** and **32**, the connecting part **3310** may be positioned on the seat part **320**, such as its armrest **3220**, through the positioning protrusion **3322**. Correspondingly, at least one positioning hole **32212** is provided on the second sliding part **32211** of the seat (in combination with FIG. **27**) for engaging with the positioning protrusion **3322**. For the arrangement of the positioning holes **32212**, a perspective view of the armrest **3220** and the second sliding part **32211** thereon is shown in FIG. **27**, which shows a positional relationship between the second sliding part **32211** and the positioning hole **32212**. For example, the positioning hole **32212** may be provided as an opening extending from one side of the second sliding part **32211**. In addition, the positioning hole **32212** may be provided as one positioning hole or a plurality of positioning holes, so that the storage device **330** may be positioned at several positions on the armrest **3220** of the seat part **320** through the selective engagement between the positioning protrusion **3322** of the positioning element **3320** and the positioning hole **32212**. When the seat is used by a smaller child, the storage device **330** may be adjusted to a backward position (i.e., closer to the backrest), so that the smaller child can easily access the objects in the storage device **330**. When the seat is used by a larger child, the storage device **330** may be adjusted to a forward position (i.e., away from the backrest), so that it is convenient for the older child to access the objects in the storage device **330**, and will not limit the movement space of the child's arms.

[0183] By providing the positioning element **3320** with the elastic arm **3321** and the positioning protrusion **3322** on the connecting part **3310**, as the elastic arm **3321** of the positioning element **3320** constantly provides a force to bias the positioning protrusion **3322** toward the seat, the positioning protrusion **3322** is engaged with the corresponding positioning hole **32212** on the connecting structure **3221** under the bias of the elastic arm **3321** when the storage device **330** slides to the appropriate position with respect to the seat, so that the storage device **330** is positioned at this position. Thus, the storage device **330** can be positioned in different usage positions as desired, further improving convenience of use and user experience.

[0184] Referring to FIGS. **33** and **34**, FIG. **33** shows a perspective view of the seat part **320** of FIG. **29** from another angle, and FIG. **34** shows an enlarged view of a part J shown in FIG. **33**. FIG. **33** shows the positioning element **3320** provided in the connecting part **3310** viewed from the other side, and when viewed from this side, at least a part of the positioning element **3320** is visible (i.e., not shielded).

[0185] As shown in FIG. **33** and FIG. **34**, in one embodiment, the positioning element **3320** further includes an operating part **3323**. In one embodiment, the operating part **3323** is located on an upper side of the other end of the positioning element **3320** (e.g., the second end **3320***b*), and is at least partially exposed. For example, an opening may be provided on a side of the armrest **3220** close to a sitting position of the child, so that the operating part **3323** is visible from the outside, and the user can contact or grasp the operating part **3323** for manual operation. By operating the operating part **3323**, the positioning element **3320** may be disengaged from the seat, so that the storage device **330** may slide to other positions with respect to the seat. For example, in one embodiment, the operating part **3323** may be pulled upward to overcome an elastic rotation of the elastic arm **3321** (that is, to overcome a biasing force provided by the elastic rotation to the positioning protrusion **3322**), so that the positioning protrusion **3322** is disengaged with the positioning hole **32212** provided on the seat part **320**. At this time, the storage device **330** may be continued to be pulled in an axial direction, so that the connecting part **3310** of the storage device **330** (such as the first sliding part as described above) slides on the armrest **3220** of the seat part **320** (such as the second sliding part as described above) to detach the storage device **330** from the seat part **320**, or adjust the storage device **330** to another position and engage with another positioning hole, thereby being suitable for children of different sizes, and further improving the operational convenience.

[0186] Through the sliding connection between the connecting part and the seat, the storage device of the present disclosure can slide forward and rearward and detachably installed on the seat, thereby restricting the movement of the storage device in the left-right direction and the vertical direction of the seat. When the child on the seat accidentally kicks the storage device from below, or when the child accidentally hits the storage device sideways while moving from one side of the seat to the seat part, the storage device according to the present disclosure can ensure the stability of the storage device and the objects placed therein, and improve the usage safety and user experience of the seat. In addition, the storage device and the seat according to the present disclosure are simple in structure and easy to be assembled and disassembled, and the storage device may be detached from the seat when it is not needed, thereby reducing the packaging volume of the seat and reducing the transportation cost.

[0187] Although relative terms such as “above” and “under” are used herein to describe the relationship of one component with respect to another component, such terms are used herein only for the sake of convenience, for example, in the direction shown in the figure, it should be understood that if the referenced device is inversed upside down, a component described as “above” will become a component described as “under”. When a structure is described as “above” another structure, it probably means that the structure is integrally formed on another structure, or, the structure is “directly” disposed on another structure, or, the structure is “indirectly” disposed on another structure through an additional structure.

[0188] The terms “a”, “an”, “the”, “said” and “at least one”, are used to express the presence of one or more the element/constitute/or the like. The terms “comprise”, “include” and “have” are intended to be inclusive, and mean there may be additional elements/constituents/or the like other than the listed elements/constituents/or the like. The “first” and “second” are used only as marks, and are not numerical restriction to the objects.

[0189] Unless otherwise defined, all terms including technical and scientific terms used herein have the same meaning as commonly understood by those ordinary skill in the art to which the present disclosure belongs. It should also be understood that a term should be interpreted as having a meaning consistent with its meaning in the context of the related art, and will not be interpreted as an idealized or overly formal meaning unless explicitly defined herein.

[0190] It should be understood that the application of the present disclosure is not limit to the detailed structure and arrangement of components provided in this specification. The present disclosure can have other embodiments, and can be implemented and carried out in various ways. The aforementioned variations and modifications fall within the scope of the present disclosure.

Claims

1. A seating assembly for detachably connecting to a backrest of a child safety seat, wherein, the seating assembly comprises: a base; a seating body slidably connected to the base, the seating body being positioned at different positions with respect to the base; and a movable element located between the backrest and the seating body, wherein when the backrest is detached with respect to the seating body, the movable element is disengaged from the backrest and locked with the base to lock movement of the seating body with respect to the base.

2. The seating assembly according to claim 1, wherein, an engaging hole is formed in the base, and when the backrest is detached with respect to the seating body, the movable element is disengaged from the backrest and is engaged with the engaging hole.

3. The seating assembly according to claim 1, wherein, an engaging hole is formed in the base, the movable element comprises a rotating rod rotatably connected to the seating body and an elastic element connected to the rotating rod, the rotating rod comprises a first end and a second end, when the backrest is not detached, the second end of the rotating rod is pressed by the backrest, so that the first end is opposite to the engaging hole and located outside the engaging hole, and the elastic element is configured to provide a force for moving the first end of the rotating rod toward the engaging hole.

4. The seating assembly according to claim 3, wherein, the backrest comprises an abutting element, and when the backrest is not detached, the abutting element abuts against the second end of the rotating rod.

5. The seating assembly according to claim 1, wherein, when the seating body slides to a preset position with respect to the base, the backrest is in a vertical state.

6. The seating assembly according to claim 1, wherein, the seating assembly comprises an adjustment assembly between the base and the seating body, the base comprises a positioning guide rail, and the positioning guide rail is provided with a plurality of positioning holes spaced apart from each other, and the seating body is slidably provided on the base and cooperates with respective one of the positioning holes through the adjustment assembly.

7. The seating assembly according to claim 6, wherein, the seating body is provided with a guide slot for accommodating the positioning guide rail, a length of the guide slot is greater than a length of the positioning guide rail, the seating body comprises a first abutting part, and the first abutting part is placed on a side of the guide slot away from the backrest to abut against the positioning guide rail when the seating body slides to a preset position.

8. The seating assembly according to claim 6, wherein, the seating body comprises a seat body, and the seat body comprises an upper shell and a lower shell connected to each other, the adjustment assembly comprises a positioning element connected between the upper shell and the lower shell, an elastic element connected to the positioning element, and an adjusting element connected to the positioning element, the positioning element is selectively placed in one of the positioning holes, and the adjusting element is movably connected to the lower shell.

9. The seating assembly according to claim 8, wherein, the positioning element comprises a positioning rod, a connecting element connected to the positioning rod, and a linking element connected between the adjusting element and the connecting element, the lower shell is formed with a first guide groove, an extending direction of the first guide groove is parallel to an axial direction of the positioning hole, the connecting element is movably placed in the first guide groove, the connecting element is formed with a receiving groove, and the elastic element is placed in the receiving groove and abuts against the connecting element.

10. The seating assembly according to claim 9, wherein, the seating body further comprises armrests on both sides of the seat body, and a connecting hole corresponding to the first guide groove is formed on an inner side of the armrest, and the positioning element passes through the

positioning hole and is placed in the connecting hole.
11. The seating assembly according to claim 1, wherein, the backrest comprises a locking element, when the seating body does not slide to a preset position with respect to the base, the locking element is not capable of being unlocked, and when the seating body slides to the preset position with respect to the base, the locking element is capable of being unlocked.
12. The seating assembly according to claim 11, wherein, a rear edge of the base is formed with an avoidance slot facing the backrest, and when the seating body slides to a preset position, the locking element is opposite to the avoidance slot, so as to allow the backrest to be detached, wherein, the locking element comprises an engaging hook and an operating element rotatably provided on the backrest, the seating assembly further comprises an engaging rod fixed on the seating body, the engaging hook is fixedly connected to the operating element, the engaging hook has a hook part facing the engaging rod and hooking the engaging rod, the engaging hook slides with respect to the base along with the seating body, and the hook part is opposite to the avoidance slot when the seating body slides to the preset position.
13. The seating assembly according to claim 11, wherein, a rear edge of the base is formed with an avoidance slot facing the backrest, when the seating body slides to a preset position, the locking element is unlocked through the avoidance slot, and when the seating body does not slide to the preset position, the locking element is not capable of being unlocked through the avoidance slot.
14. The seating assembly according to claim 13 wherein, the locking element comprises an engaging hook and an operating element rotatably provided on the backrest, the seating assembly further comprises an engaging rod fixed on the seating body, the engaging hook is fixedly connected to the operating element, the engaging hook has a hook part facing the engaging rod and hooking the engaging rod, the engaging hook slides with respect to the base along with the seating body, and the hook part is opposite to the avoidance slot when the seating body slides to the preset position.
15. The seating assembly according to claim 6, wherein, the seating body comprises a seat body, the seat body is formed with an accommodating groove for receiving the locking element, and the locking element is fixed to a bottom end of the backrest.
16. The seating assembly according to claim 1, wherein, the seating body includes an armrest, a top portion of the armrest is formed with a first curved sliding slot and a second curved sliding slot communicated with each other, the backrest comprises a leaning part and side leaning parts on both sides of the leaning part, a sliding part protrudes from a bottom portion of each of the side leaning parts, the sliding part is slidably placed in the first curved sliding slot, a cross-section of each of the first curved sliding slot and the second curved sliding slot is matched with a cross-section of the sliding part.
17. The child safety seat according to claim **49**, wherein the folding mechanism is configured to fold a backrest of a child safety seat with respect to a seat part, and wherein, the folding mechanism comprises: a sliding groove disposed on one of the seat part and the backrest; a sliding rib disposed on the other of the seat part and the backrest; a locking structure operated to switch between a locking position and an unlocking position, wherein the backrest is fixed with respect to the seat part and is in an unfolded state when the locking structure is in the locking position; and the backrest is foldable with respect to the seat part by means of sliding fit between the sliding groove and the sliding rib to be in a folded state when the locking structure is in the unlocking position.
18. The child safety seat according to claim 17, wherein, the sliding groove is disposed on each of both sides of the seat part, and the sliding rib is disposed on each of both sides of the backrest, wherein the sliding groove is disposed on a top surface of an armrest at each of both sides of the seat part, the armrest at each of both sides of the seat part comprises a front armrest and a rear armrest spaced apart from each other; in an unfolded state, the sliding rib is in a rear sliding groove of the rear armrest, and in a folded state, the sliding rib is in a front sliding groove of the front armrest.

19-22. (canceled)

23. The child safety seat according to claim 17, wherein, the locking structure is disposed at a lower side of the backrest; the folding mechanism further comprises an engaging rod fixed to the child safety seat, and the locking structure is in the locking position by the engagement with the engaging rod and is in the unlocking position by the disengagement from the engaging rod.

24-48. (canceled)

49. A child safety seat, wherein, the child safety seat comprises: the seating assembly according to claim 1; and a folding mechanism, allowing a backrest of the child safety seat to be folded or selectively separated with respect to the seating body.